

Arjun Shanmugam

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EDUCATION

Brown University

Providence, RI | May 2023

B.S. in Computer Science-Economics

GPA: 3.96/4.00; 4.00 within major. **GRE:** 170/170 Q; 166/170 V; 5/6 W

Relevant Coursework: Advanced Topics in Econometrics, Bayesian Statistics, Big Data, Computational Linguistics (NLP), Computational Probability and Statistics, Deep Learning, Differential Equations, Discrete Mathematics, Linear Algebra, Machine Learning, Mathematical Econometrics, Multivariable Calculus, Probability, Real Analysis, Software Engineering, Statistical Inference
Skills: Causal inference, empirical Bayes, experimentation, SciPy stack, SQL, statistics, supervised machine learning with big data

EXPERIENCE

Uber Technologies, Inc.

Scientist, Tech

New York, NY | April 2026 – Present

- Used machine learning methods to model the incremental impact of incentives for different types of drivers, allowing more efficient targeting of ~\$400 million in yearly incentive spend
- Devised and productionized a time series model to detect anomalies in Uber's driver onboarding funnel, preventing incidents that previously reduced driver onboarding flow in the US by 20%
- Engineered an LLM-powered RAG system for experiment readouts, saving time for scientists and engineers by making past experiment results queryable using natural language

Applied Scientist, Marketing

New York, NY | April 2025 – April 2026

- Devised a novel Bayesian shrinkage algorithm for two noisy surveys, reducing MSE in estimates of their correlation by 40% compared to classical measurement error correction
- Engineered and productionized an empirical Bayes smoothing procedure for noisy experimental estimates, significantly increasing reporting accuracy and enabling non-technical stakeholders to interpret results with higher confidence
- Developed a robust framework to estimate marketing spend elasticities, validated through out-of-sample testing
- Designed and ran complex experiments to test various marketing strategies, driving funds towards more efficient campaigns
- Explained difficult statistical concepts to non-technical stakeholders, convincing them to adopt new best practices

Opportunity Insights at Harvard University

Pre-doctoral Fellow under Professors Raj Chetty, John Friedman, and Nathan Hendren

Cambridge, MA | June 2023 – April 2025

- Estimated the causal effect of elite college attendance using restricted admissions records and IRS tax data for "Diversifying Society's Leaders?" [covered by The New York Times](#)
- Performed calculations to estimate the joint distribution of SAT score and parent income, [covered by The New York Times](#)
- Applied statistical tools such as inverse propensity score reweighting and instrumental variables to causal questions
- Built a Python and SQL codebase that processes the universe of raw American tax forms into a 12 billion row database ("The IRS Databank") used by the IRS and Treasury, reducing build time from one month to 3 days
- Transitioned lab's big data and computing infrastructure from SAS to Python and SQL, resulting in huge efficiency gains

Opportunity Insights at Harvard University

NLP Research Assistant to Professors John Friedman, Amy Handlan, Nathan Hendren

Cambridge, MA | May 2022 – September 2022

- Examined differences in language used by male vs. female reviewers at the *Journal of Public Economics*
- Engineered NLP pipeline using pandas and Scikit-Learn to extract features from manuscripts and reviewer reports
- Trained and tested cross-validated LASSO machine learning models on document term matrices extracted from text
- Devised an algorithm which separates papers' introductions from the rest of their text with over 70 percent accuracy
- Formulated parametric and nonparametric models of referee report text, each controlling for paper content in different ways
- Parallelized computationally intensive processes such as the estimation of likelihood ratios using joblib

PROJECTS

[Eviction and Crime: Quasi-Experimental Evidence from Boston](#)

Author

Providence, RI | September 2022 – Present

- Won Outstanding Honors Thesis Award in Economics (best undergraduate economics thesis at Brown)
- Spatially joined every attempted eviction in Boston since 2019 with 1 million crime incidents using computing cluster
- Estimated propensity scores using maximum likelihood estimation to balance treatment and control group on observables
- Estimated effect of eviction on crime around the property with staggered, doubly robust difference-in-difference design

LEADERSHIP

Brown Department of Economics

Teaching Assistant, Mathematical Econometrics, Professor Jonathan Roth

Providence, RI | January 2022 – May 2023

- Taught econometric theory and causal inference to 60 students, earning top reviews (mean 4.8/5, median 5/5)
- Received strong feedback: "Arjun is single handedly the most fantastic TA I have had while at Brown. His deep understanding of topics and his willingness and availability to explain difficult concepts made him more than just a TA!"